ADITYA PAWAR

Boston area | GitHub | LinkedIn | Google Scholar | Website

PROFESSIONAL SUMMARY

Data scientist and ML/AI platform engineer with 8+ years building data platforms, pipelines and, more recently, agentic AI for translational scientists. At Fred Hutch I architected and own a multi-cohort translational AI platform that unifies metastatic prostate WGS, longitudinal ctDNA under 177Lu-PSMA radioligand therapy, and multi-omics bladder cancer data under one provenance-tracked data model, with grounded agents that answer only through validated queries. Technical lead on DRAGNX, a pan-cancer generative model (manuscript in revision). Deep expertise in cfDNA/ctDNA, liquid biopsy, WGS/WES and transcriptomics. I build every component with tests, versioning, audit trails and documentation so the owning team can run it. Mentored interns.

TECHNICAL SKILLS

Agentic AI and LLM systems: tool-calling agents (JSON-Schema-typed tools), retrieval and grounding, number-verification guardrails, multi-model routing across frontier and local open-weights models, prompt caching, evaluation harnesses, audit and usage telemetry, sub-agents, MCP integrations, multimodal vision-language agents, LLM fine-tuning and inference

Data platforms and governance: PostgreSQL schema and entity modeling, data versioning and lineage, staging-to-curated ETL with independent consistency checks, Parquet/DuckDB, Databricks, metadata harmonization, access control (institutional SSO)

Deployment and engineering: Docker, Apptainer, AWS (S3, Batch, EC2, Lambda, SageMaker), Kubernetes, Terraform, GitHub Actions CI/CD, Nextflow DSL2, Snakemake, SLURM/HPC, MLflow, REST APIs, Streamlit, Flask

Translational genomics: cfDNA/ctDNA, liquid biopsy, WGS/WES, EM-seq methylation, RNA-seq, single-cell, SNV/SV callers, copy number (TITAN, ichorCNA), ANNOVAR, ACMG germline annotation, NVIDIA Parabricks

Machine learning: PyTorch, TensorFlow, Hugging Face, autoencoders, GNNs, CNNs/Transformers, XGBoost, survival models, CUDA, multi-GPU and mixed-precision training

Languages: Python, SQL, R, Bash, Java

PROFESSIONAL EXPERIENCE

Data Scientist, ML and AI Platform Engineering, Fred Hutchinson Cancer Center, Gavin Ha Lab
Seattle, WA | June 2025 to present

Translational AI platform (architect and owner)

- Built and own a multi-cohort translational AI platform for the Ha Lab and collaborating labs, designed for both computational and wet-lab scientists. It unifies three cancer programs:
  - a metastatic prostate rapid-autopsy WGS cohort
  - a longitudinal ctDNA cohort of mCRPC patients sampled before, during and after 177Lu-PSMA-617 (Pluvicto) radioligand therapy
  - a multi-omics metastatic bladder cancer cohort (WGS/WES, bulk RNA-seq, cfDNA)
- Common data model and biomarker ontology: a PostgreSQL registry modeling subjects, cohorts, treatment timepoints, specimens (tissue, autopsy, plasma), assays, tumor-normal pairs, analysis runs and results, with controlled vocabularies for biomarker definitions such as ctDNA tumor fraction, clonal status (founder, shared, private) and caller consensus.
- Versioned biomarker data products: SNV, SV, copy-number, ctDNA tumor-fraction and expression data are released as validated Parquet/DuckDB builds. Queries over large variant tables return in under a second, and outputs reproduce the lab's reference results exactly. Extending to single-cell RNA-seq.
- Grounded ctDNA and genomics agents: deterministic rules answer common questions first, and an LLM agent answers the rest only through schema-validated tools.
- Compliant multi-model gateway and telemetry: agents run through an approved, contract-covered model gateway, with model choice per question (fast or strong) and switching mid-conversation, plus a self-hosted open-weights model on an on-premises GPU as the fallback. An audit log records each question's tools and tokens, and prompt caching is enabled.

Machine learning and agentic curation

- Technical lead, DRAGNX: built a pan-cancer generative model trained on a large corpus of RNA-seq samples (autoencoders), with a connector that maps cfDNA and tumor EM-seq methylation to gene expression, using multi-GPU mixed-precision training on NVIDIA L40 GPUs. Manuscript in revision.
- Agentic curation: built a multimodal (text and image) agent that curates TITAN copy-number outputs, reducing manual review from 40 hours to 15 minutes per batch, and extended it into a dual-model design (two frontier models) scored against one shared curation skill and benchmarked against TITAN's own optimal solution.

Pipelines, GPU acceleration and clinical genomics

- Shared pipelines: converted the lab's genomics workflows into containerized Nextflow DSL2 pipelines on SLURM HPC and AWS, with GPU (L40S) execution using NVIDIA Parabricks, cutting runtime from 1 week to 12 hours (14x).

- Algorithms: added GPU-accelerated joint multi-sample calling to ichorCNA (1k+ citations) and new features to TITAN (400+ citations), collaborating toward inclusion of both in Parabricks.
- Clinical genomics: led genomic analysis across all cohorts of the longitudinal Pluvicto ctDNA study (SNVs, SVs, haplotypes, comut plots), supporting an AACR 2026 abstract and the manuscript revisions.

Senior Data Science Consultant, The James Cancer Center, Ohio State University
Columbus, OH | March 2024 to May 2025

- Built an elastic-net Cox proportional hazards model to identify gemcitabine-resistance genes, and processed genomic data on AWS with WES/RNA-seq Nextflow pipelines on AWS Batch.
- Applied scGPT to study cellular perturbations (92% accuracy on unseen cell populations) and characterized patient-derived xenograft models (copy number, SNVs, SVs).

Data Scientist II, Target Discovery Platform, NextRNA Therapeutics
Cambridge, MA | January 2022 to September 2023

- Helped build the startup's cloud bioinformatics infrastructure on Nextflow and MLOps (AWS SageMaker, MLflow, Docker, GitHub Actions CI/CD), cutting pipeline run time by 40%.
- Built CNN, Transformer and GNN models for RNA-protein interaction prediction from eCLIP data (87% AUC) and a CRISPR pooled-screen pipeline to nominate lncRNA targets.
- Built an RDKit and GNN screening system that reduced compound libraries by 60% while preserving diversity, and worked daily with protein, chemistry and biology teams.

Bioinformatics Software Engineer, Memorial Sloan Kettering Cancer Center
New York, NY | January 2021 to January 2022

- Extended Pathoman, MSK's Python/Flask germline variant annotation tool (ACMG guidelines), built a daily database-update job from multiple external sources and improved classification (45% fewer false negatives).
- Analyzed a ~28,000-patient cohort (EHR and omics) in Databricks/Spark (40% faster processing) and ran WES/WGS pipelines on 4,000+ samples on HPC.

Machine Learning Engineer, Massachusetts Institute of Technology (MIT)
Cambridge, MA | May 2020 to December 2020

- Built deep learning models for protein interaction and antibody-antigen binding (CNN-LSTM 85% ROC-AUC, dilated CNN with attention 87%), deployed on AWS SageMaker with MLflow, and cut training time from 72 to 8 hours.

Data Scientist Intern, PureTech Health
Boston, MA | May 2019 to August 2019

- Built tools to process bulk RNA-seq and supported single-cell sequencing pipelines.

Data Scientist Co-op, Surface Oncology
Cambridge, MA | June 2018 to December 2018

- Built Shiny applications for biomarker and target discovery and a single-cell sequencing pipeline on AWS.

SELECTED PUBLICATIONS

- Adil, Hanratty, **Pawar**, et al. Decoding gene expression from cell-free DNA methylation profiles. Manuscript in revision.
- Prostate cancer phenotypes determined from circulating tumor DNA associate with outcomes to 177Lu-PSMA-617 therapy. AACR Annual Meeting 2026, late-breaking abstract.
- Patton, McDeed, Netzley, **Pawar**, et al. Gene expression inference from cell-free DNA using uncertainty-aware deep learning. bioRxiv preprint, 2026.

EDUCATION

Master of Science in Applied Mathematics, Northeastern University, Boston, MA | 2017 to 2019